KR-ACC-24: Minor BGP Route Churn and OSPF Adjacency Loss on Arista Access Switches

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1093

Description

This Standard Operating Procedure (SOP) is designed to address a minor failure domain on the KR-ACC-24 node at the KR Puram Access site. The failure domain comprises BGP route churn and OSPF adjacency loss on Arista access switches. This phenomenon occurs due to the dynamic nature of routing protocols, particularly BGP and OSPF, which are designed to adapt to changing network topologies and routing tables. In this scenario, the Arista access switch's BGP and OSPF processes are experiencing a high rate of route churn, leading to OSPF adjacency loss. This, in turn, causes the switch to flood ISIS LSPs, resulting in the ALARM_1093 alarm. The underlying protocol mechanics driving this failure domain are the periodic refresh of BGP routes and the re-computation of OSPF adjacencies in response to changes in the network topology.

Pre-requisite Checks

- Perform a site visit to KR Puram Access to visually inspect the KR-ACC-24 node and its connections.
- Verify that the redundant link status is operational and functioning as expected.
- Confirm that the Arista access switch is running the latest software version.

Potential Root Causes

Cause	Probability
High BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP router identifier 10.10.10.10, local AS number 65001, Description: This command displays the BGP router identifier and the local AS number, which are essential for identifying the BGP process on the Arista access switch.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the OSPF neighbor relationships, including the neighbor ID, priority, state, and dead time, which are critical in identifying OSPF adjacency loss.

Command: show isis database, Expected Output: ISIS Level 2 Database, Description: This command displays the ISIS database, which includes the LSPs and their associated information, such as the LSP ID,
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sequence number, and age.
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Resolution Steps

Step 1: Step: Reboot the Arista Access Switch, Command: reload, Description: Rebooting the Arista access switch can resolve the issue by resetting the BGP and OSPF processes and re-establishing OSPF adjacencies.

Step 2: Step: Verify BGP Route Churn, Command: show bgp summary, Description: Verify that the BGP route churn has been resolved by checking the BGP summary output for any changes in the BGP router identifier or the local AS number.

Step 3: Step: Verify OSPF Adjacency Loss, Command: show ospf neighbor, Description: Verify that the OSPF adjacency loss has been resolved by checking the OSPF neighbor relationships for any changes in the neighbor ID, priority, state, or dead time.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP router identifier 10.10.10.10, local AS number 65001, Description: Verify that the BGP route churn has been resolved by checking the BGP summary output for any changes in the BGP router identifier or the local AS number.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify that the OSPF adjacency loss has been resolved by checking the OSPF neighbor relationships for any changes in the neighbor ID, priority, state, or dead time.
```

Roll-back Procedures

- Step: Revert the Arista Access Switch to its Previous Configuration, Command: load merge bootflash:config_backup, Description: If the fix fails, revert the Arista access switch to its previous configuration by loading the backup configuration file.

- Step: Reboot the Arista Access Switch, Command: reload, Description: Reboot the Arista access switch to ensure that the changes are fully applied.